
Information

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Outline

- Information
- Entropy
- Cross Entropy
- Error Backpropagation Learning
- Mutual Information
- Kullback Leibler Divergence
- Independent Component Analysis (ICA)
- Learning for ICA
- Blind Source Separation

Information

- Discrete random variable X is defined in the sample set Ψ

$$\Psi = \{x_k | k = 0, \pm 1, \dots, \pm K\}$$

- Event $X = x_k$ occurs with probability $p_k = P(X = x_k)$

- **Information** $\equiv$ surprise $\equiv$ uncertainty

The amount of information of the event is related to the *inverse* of the probability of occurrence. That is, the lower the probability p_k is, the more “surprise” there is, and the more “information”.

$$I(x_k) = \log\left(\frac{1}{p_k}\right) = -\log p_k$$

내일도 지구가 회전한다

$p_k = 1$: 정보($\times$), surprise($\times$)

내일 미국이 북한을 공격한다

$p_k \ll 1$: 정보(0), surprise(0)

Information

- base=2 $\Rightarrow$ 정보단위 bits
- base=e $\Rightarrow$ 정보단위 nats
- 32 bit : 한 code의 정보는 $I(x_k) = -\log(\frac{1}{2^{32}}) = 32$

- ① $I(x_k) = 0$ for $p_k = 1$
- ② $I(x_k) \geq 0$ for $0 \leq p_k \leq 1$
- ③ $I(x_k) \geq I(x_i)$ for $p_k \leq p_i$

- **Entropy** : a measure of the *average amount of information conveyed per message*, i.e., expectation of Information

$$H(X) = E[I(X)] = \sum_{k=-K}^K p_k I(x_k) = - \sum_{k=-K}^K p_k \log p_k$$

Information

- Maximum entropy : when p_k is equiprobable.

$$0 \leq H(X) \leq - \sum_{k=-K}^K \frac{1}{2K+1} \log \frac{1}{2K+1} = \log(2K+1)$$

- $H(X) = 0$ for an event that $p_k = 1$ o/w $p_k = 0$
- Theorem (Gray 1990)

$$\sum_k p_k \log\left(\frac{p_k}{q_k}\right) \geq 0$$

- Relative entropy (or Kullback – Leibler divergence)

$$D_{p\|q} = \sum_{x \in X} p_X(x) \log\left(\frac{p_X(x)}{q_X(x)}\right)$$

where $p_X(x)$ is probability mass ftn.(pmf), $q_X(x)$ is reference pmf

Information

- Relative entropy (or Kullback – Leibler divergence) for neural network

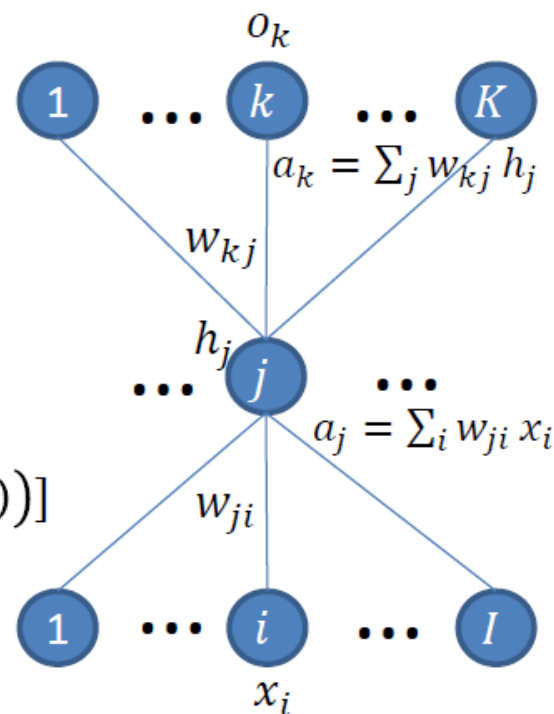
$$D_{p||q} = \sum_{x \in X} p(x) \log \left(\frac{p(x)}{q(x; W)} \right) = \sum_{x \in X} p(x) \log p(x) - \sum_{x \in X} p(x) \log q(x; W)$$

- Cross entropy for one-hot classification by deep learning

$$C_{p||q}(x; W) = - \sum_x \sum_k p_k(x) \log q_k(x; W)$$

- Cross entropy for multi-label classification by deep learning

$$C_{p||q}(X; W) = - \sum_x \sum_k [p_k(x) \log p_k(x; W) + (1 - p_k(x)) \log(1 - p_k(x; W))]$$



Backpropagation Learning Rule

- Empirical Risk Function:

$E_d(w)$

Regression: L_2 , linear

01001101: cross-entropy, sigmoid

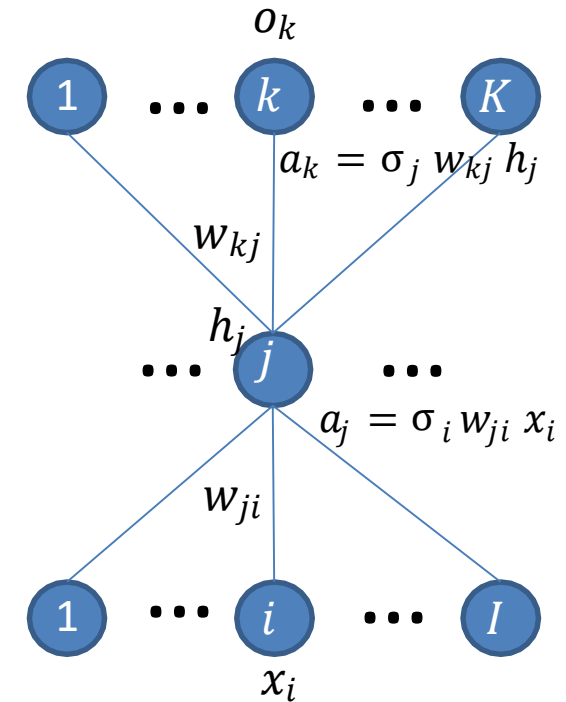
00001000: cross-entropy, soft-max

- Gradient descent for **output layer**:

$$\Delta w_{kj} = -\eta \frac{\partial E_d}{\partial w_{kj}}$$

- Chain rule:

$$\frac{\partial E_d}{\partial w_{kj}} = \frac{\partial E_d}{\partial a_k} \frac{\partial a_k}{\partial w_{kj}} = \frac{\partial E_d}{\partial a_k} h_j$$



Backpropagation Learning Rule

- For multi-label classification (ex, output: 0110100), sigmoid activation function is used and the loss is defined by the cross entropy loss function:

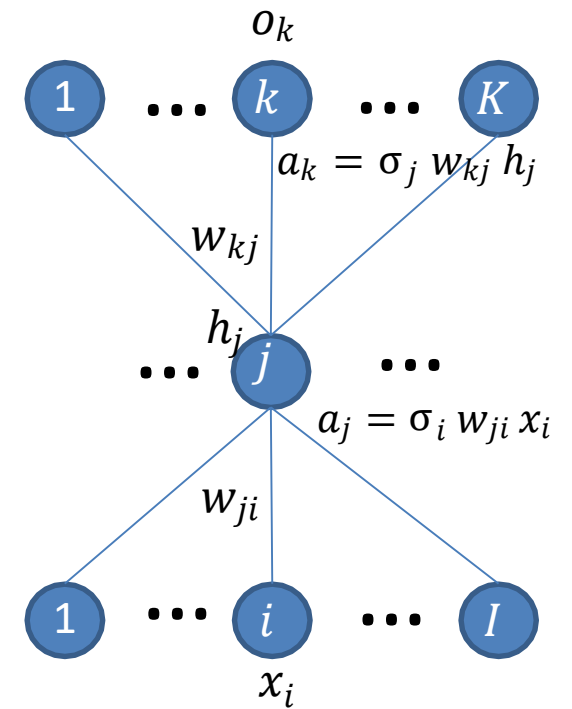
$E(w) = -\sigma_k^K [t_k \log o_k(x, w) + (1 - t_k) \log(1 - o_k(x, w))]$, where

$$o_k = \sigma(a_k) = \frac{1}{1 + e^{-a_k}}. \text{ Then find } \frac{\partial E}{\partial a_k}.$$

Sol.)

$$\frac{\partial E_d}{\partial w_{kj}} = \frac{\partial E_d}{\partial a_k} \frac{\partial a_k}{\partial w_{kj}} = \frac{\partial E_d}{\partial a_k} h_j$$

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$$\frac{\partial E}{\partial a_k} = -t_k \frac{1}{o_k} \frac{\partial o_k}{\partial a_k} - (1 - t_k) \frac{-1}{1 - o_k} \frac{\partial o_k}{\partial a_k}$$

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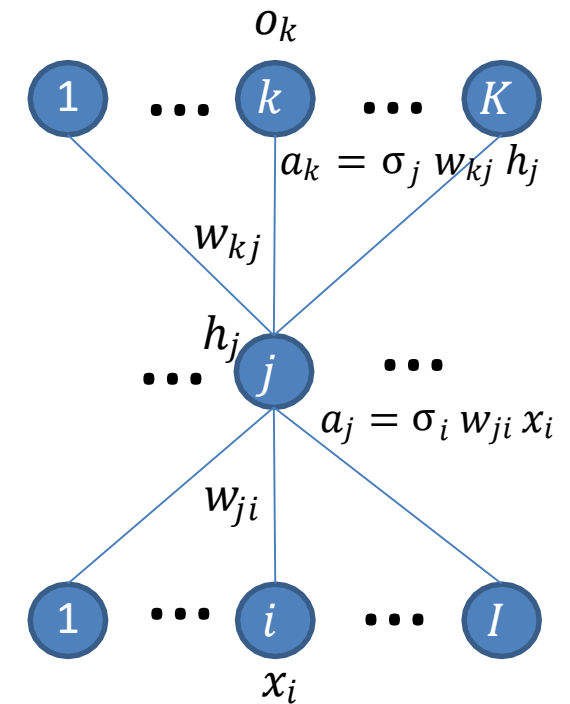
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$$\frac{\partial E_d}{\partial w_{kj}} = \frac{\partial E_d}{\partial a_k} \frac{\partial a_k}{\partial w_{kj}} = \frac{\partial E_d}{\partial a_k} h_j$$

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Backpropagation Learning Rule

- For multi-class classification (ex, [0 0 0 1 0 0]), the softmax activation function is used and the loss is defined by the cross entropy loss function: $E(w) = -\sum_i^K t_i \log(o_i(x, w))$, where $o_k(x, w) = \frac{e^{a_k}}{\sum_j e^{a_j}}$. The target value $t_k \in \{0, 1\}$ is labelled by 1 hot vector. Then find $\frac{\partial E}{\partial a_k}$.

Sol.)

$$\frac{\partial E_n}{\partial a_k} = \frac{\partial}{\partial a_k} \left(-\sum_i^K t_i \log\left(\frac{e^{a_i}}{\sum_j e^{a_j}}\right) \right)$$

Backpropagation Learning Rule

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Backpropagation Learning Rule

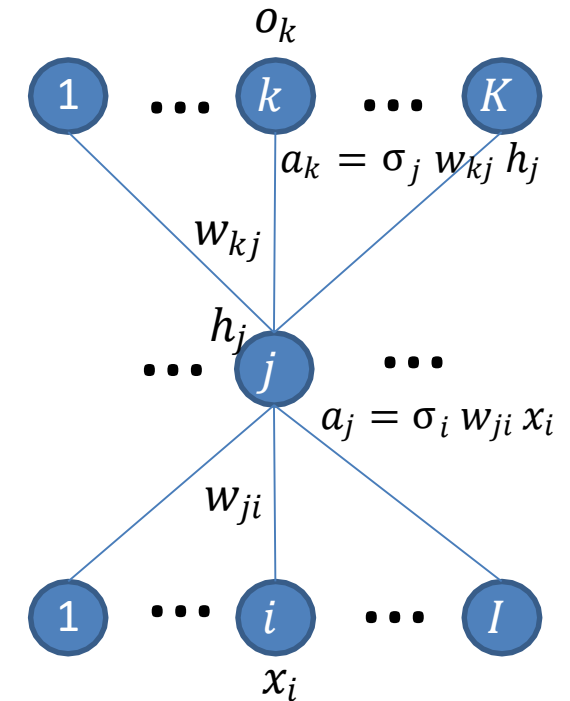
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 &= -t_k + \frac{e^{a_k}}{\sum_j e^{a_j}} \sum_i t_i = o_k - t_k = -(t_k - o_k) = -\delta_k
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$$\frac{\partial E_d}{\partial w_{kj}} = \frac{\partial E_d}{\partial a_k} \frac{\partial a_k}{\partial w_{kj}} = \frac{\partial E_d}{\partial a_k} h_j$$

$$\Delta w_{kj} = -\eta \frac{\partial E_d}{\partial w_{kj}} = \eta \delta_k h_j$$



Backpropagation Learning Rule

- Empirical Risk Function:

$E_d(w)$

Regression: L_2 , linear

01001101: cross-entropy, sigmoid

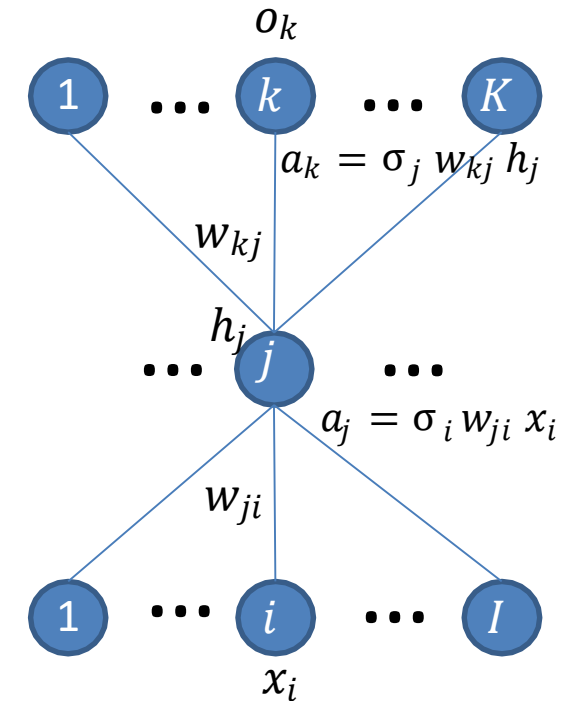
00001000: cross-entropy, soft-max

- Gradient descent for **hidden layer**:

$$\Delta w_{ji} = -\eta \frac{\partial E_d}{\partial w_{ji}}$$

- Chain rule:

$$\frac{\partial E_d}{\partial w_{ji}} = \frac{\partial E_d}{\partial a_j} \frac{\partial a_j}{\partial w_{ji}} = \frac{\partial E_d}{\partial a_j} x_i$$



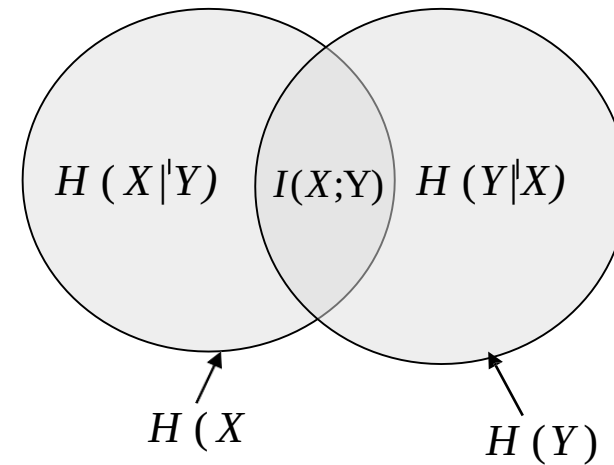
Mutual Information

- Conditional Entropy (조건부 불확실성의 량)
Y 가 관측되고 난 후의 X 의 정보기대치 (Entropy)
Y 와 연관이 있는 X 의 정보는 제외

- Theorem (Gray 1990)
 $H(X|Y) = H(X, Y) - H(Y)$
 $0 \leq H(X|Y) \leq H(X)$

- Joint Entropy
 $H(X, Y) = - \sum_{x \in X} \sum_{y \in Y} p(x, y) \log p(x, y)$

└──────────┴──────────┘
Joint probability mass function



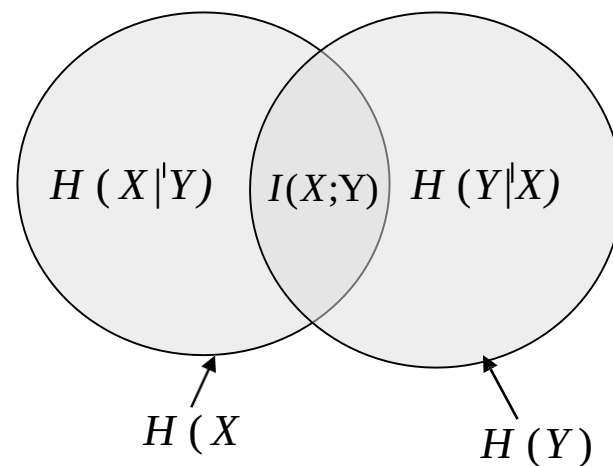
Mutual Information

- Mutual Information: Output Y 의 관측에 의해 알 수 있는 X 의 uncertainty (정보)

$$\begin{aligned} I(X; Y) &= H(X) - H(X|Y) \\ &= H(X) + H(Y) - H(X, Y) \\ &= -\sum_{x \in X} p(x) \log(p(x)) - \sum_{y \in Y} p(y) \log(p(y)) \\ &\quad + \sum_{x \in X} \sum_{y \in Y} p(x, y) \log(p(x, y)) \\ &= \sum_{x \in X} \sum_{y \in Y} p(x, y) \log\left(\frac{p(x, y)}{p(x)p(y)}\right) \end{aligned}$$

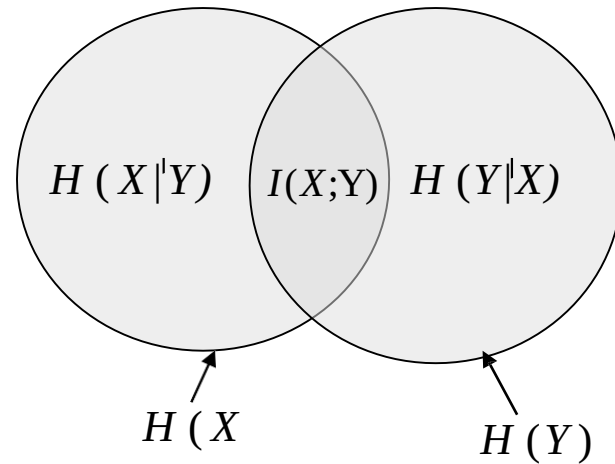
- KL-divergence & Independence ?

$$H(X) = I(X, X)$$



Mutual Information

- Properties of $I(X, Y)$
 - ① $I(Y; X) = I(X; Y)$
 - ② $I(X; Y) \geq 0$
 - ③ $I(X; Y) = H(Y) - H(Y|X)$



Mutual Information

- Mutual Information for Continuous Random Variables

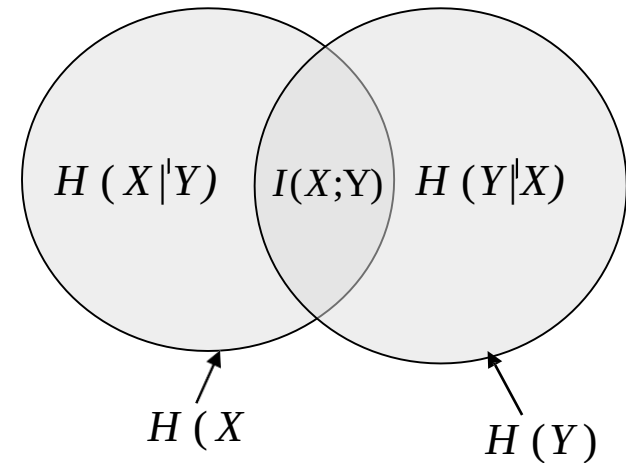
$$I(X;Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) \log \left(\frac{f_{X,Y}(x,y)}{f_X(x)f_Y(y)} \right) dx dy$$

$$I(X;Y) = h(X) - h(X|Y) = h(Y) - h(Y|X)$$

$$= h(X) + h(Y) - h(X,Y)$$

$$I(X;Y) = I(Y;X)$$

$$I(X;Y) \geq 0$$



Exercise

- 프리미어리그에서 아스널과 토트넘이 경기를 하고 있다. TV를 보며 마음껏 떠들 수 있도록 자리가 마련된 치킨집의 식객 30명과 바로 옆 삼겹살 집 식객 60명이 응원전을 펼치고 있다. 치킨집 사람들에게 어느 팀을 응원하는지 물었을 때 토트넘 10명, 아스널을 20명이 응원한다고 답했다. 삼겹살 집에서는 각 팀을 몇 명이 응원하고 있는지 확인하지 못했다.
- 치킨 집에서 '토트넘'을 응원한다는 답변에 담긴 정보량(Information Gain)은?

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- 일목요연하게 내용 정리.

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- 일목요연하게 내용 정리.

- $Information = -\log p(X = x)$

	치킨집	삼겹살집
토트넘 응원자	10	n 명
아스널 응원자	20	$(60 - n)$ 명

Exercise

- 프리미어리그에서 아스널과 토트넘이 경기를 하고 있다. TV를 보며 마음껏 떠들 수 있도록 자리가 마련된 치킨집의 식객 30명과 바로 옆 삼겹살 집 식객 60명이 응원전을 펼치고 있다. 치킨집 사람들에게 어느 팀을 응원하는지 물었을 때 토트넘 10명, 아스널을 20명이 응원한다고 답했다. 삼겹살 집에서는 각 팀을 몇 명이 응원하고 있는지 확인하지 못했다.
- 치킨 집에서 '토트넘'을 응원한다는 답변에 담긴 정보량(Information Gain)은?

- 일목요연하게 내용 정리.

- $Information = -\log P(x)$

$$P(X = \text{토트넘} | Y = \text{치킨집}) = 1/3$$

	치킨집	삼겹살집
토트넘 응원자	10	n 명
아스널 응원자	20	$(60 - n)$ 명

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- 치킨집에서 토트넘 응원하는 경우를 $X = 0$, 아스널 응원하는 경우를 $X = 1$ 이라 할 때
우측 표가 지닌 엔트로피는?

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- Entropy: $H(X) = -\sum_{x \in X} p(x) \log p(x)$

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- Entropy: $H(X) = -\sum_{x \in X} p(x) \log p(x)$

- $H(x|Y = 0) = -\sum_x p(x|Y = 0) \log p(x|Y = 0)$

- $H(x|Y = 0) = -\frac{1}{3} \log \frac{1}{3} - \frac{2}{3} \log \frac{2}{3}$

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Exercise

- KL-Divergence의 의미를 생각할 때 각 음식점 에서 두 팀을 응원할 확률분포 간의 KL-divergence, 즉 $D_{P(X|Y=0) || P(X|Y=1)}$ 을 최소로 하는 n 값을 구하시오.

$D_{P(Y=0) || P(Y=1)}$ 을 최소로한다는 것은 각 음식점에서 두팀을 응원할 확률 분포가 같게 된다는 의미이다.

즉, $P(X|Y = 0) = P(X|Y = 1)$

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즉, $P(X|Y = 0) = P(X|Y = 1)$

$$\frac{1}{3} = \frac{n}{60}, \quad \frac{2}{3} = \frac{60 - n}{60} \rightarrow n = 20$$

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Exercise

- $D_{P(X|Y=0)||P(X|Y=1)}$ 을 최소로 하는 n 값을 최적화 방법으로 구하시오.

$$n^* = \operatorname{argmin}_n D_{P(X|Y=0)||P(X|Y=1)}$$

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$$n^* = \operatorname{argmin}_n D_{P(X|Y=0)||P(X|Y=1)}$$

$$\begin{aligned} D_{P(X|Y=0)||P(X|Y=1)} &= \sum_x P(X = x|Y = 0) \log \frac{P(X = x|Y = 0)}{P(X = x|Y = 1)} \\ &= \frac{1}{3} \log \frac{\frac{1}{3}}{\frac{1}{60}} + \frac{2}{3} \log \frac{\frac{2}{3}}{\frac{2}{60-n}} \end{aligned}$$

Exercise

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 \end{aligned}$$

$$\frac{d}{dn} D_{P(X|Y=0) || P(X|Y=1)} = \frac{n}{60} \left(-\frac{20}{n^2} \right) + \frac{60-n}{60} \left(\frac{40}{(60-n)^2} \right) = -\frac{1}{3n} + \frac{2}{3(60-n)} = \frac{-60+3n}{3n(60-n)} = 0 \rightarrow n = 20$$

Exercise

- $D_{P(X|Y=0)||P(X|Y=1)}$ 을 이용하여 구한 n 이 참값이라고 할 때, 위 표가 지닌 응원팀(X)과 음식점(Y)에 관한 Mutual Information $I(X, Y)$ 을 수식을 사용하지 않고 개념적으로 구하시오. 그리고 수식을 사용하여 구하여 개념적으로 구한 경우와 비교하시오.

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응원팀과 음식점은 서로 독립이다. 그 이유는 음식점에 따라 두 팀을 응원하는 확률 분포가 달라지지 않기 때문이다. 따라서 Mutual Information은 0 이다.

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$$\begin{aligned}
 I(X, Y) &= \sum_{x,y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)} = \sum_{x,y} p(x,y)p(y) \log \frac{p(x|y)p(y)}{p(x)p(y)} \\
 &= \frac{1}{3} \frac{1}{3} \log \frac{\frac{11}{33}}{\frac{11}{33}} + \frac{2}{3} \frac{1}{3} \log \frac{\frac{21}{33}}{\frac{21}{33}} + \frac{1}{3} \frac{2}{3} \log \frac{\frac{12}{33}}{\frac{12}{33}} + \frac{2}{3} \frac{2}{3} \log \frac{\frac{22}{33}}{\frac{22}{33}} = 0.
 \end{aligned}$$

Exercise

- Mutual Information과 Conditional Entropy의 관계에 의하여 $H(X|Y)$ 을 구하시오.

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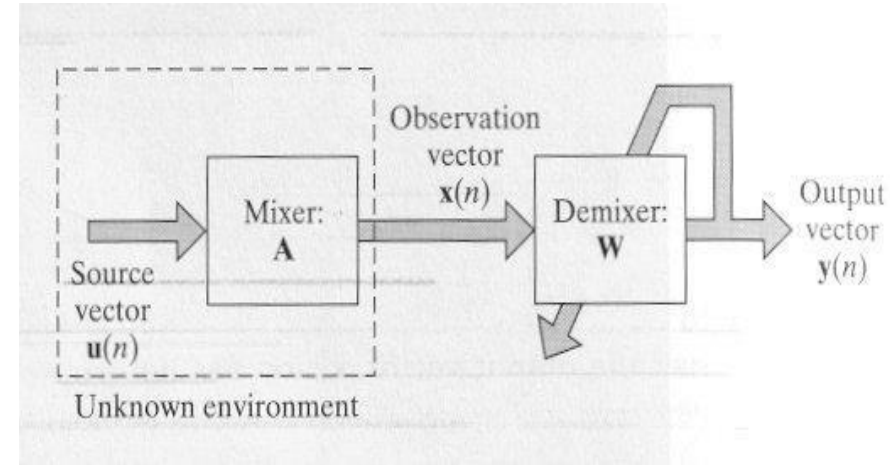
$$I(X, Y) = H(X) - H(X|Y) = 0$$

$X \setminus Y$	치킨집	삼겹살집
토틸넘 응원자	10	n 명
아스널 응원자	20	$(60 - n)$ 명

$$H(X|Y) = H(X) = -\sigma_x p(x) \log p(x) = -\frac{1}{3} \log \frac{1}{3} - \frac{2}{3} \log \frac{2}{3}$$

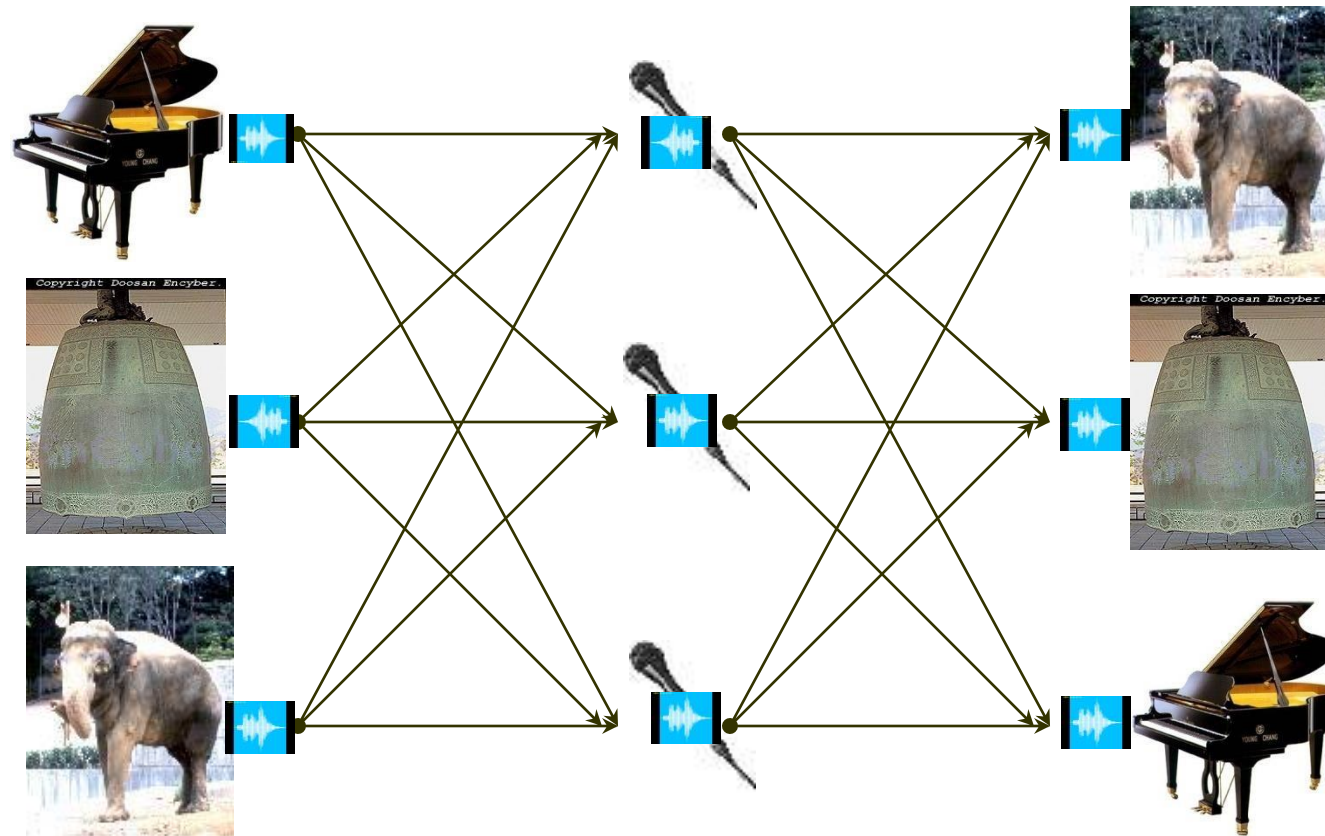
ICA(Independent Component Analysis)

- **Blind source separation** problem:
Given N independent realizations of the observation vector X , find an estimate of the inverse of the mixing matrix A
- Algorithm of ICA:
 - “as statistically independent as possible”
 - minimizing the mutual information between the each components of the output vector .



ICA(Independent Component Analysis)

- ICA Example



ICA(Independent Component Analysis)

- blind source separation problem

$U = [u_1, u_2, \dots, u_m]^T$: Independent Sources

$X = AU$, A : Mixing Matrix

$X = [x_1, x_2, \dots, x_m]^T$: Observations

$Y = WX$, W : Demixing Matrix

U, X, Y : Zero mean Signals

- $Y = WX = WAU = DPU$,
where D : Diagonal matrix, P : Permutation matrix
- How to find W ?

ICA(Independent Component Analysis)

- ICA : statistical independence
- Applications
 - Speech separation : teleconference
 - Array antenna processing
 - Multisensor biomedical records
(태아의 심장박동을 어머니 심장박동과 분리)
 - Financial market data (Dominant data 추출)
 - Feature Extraction

ICA(Independent Component Analysis)

- Criterion for Statistical Independence

Goal : Y_i, Y_j 간 mutual information을 최소화

$$\min I(Y_i; Y_j) \quad i, j = 1, \dots, m$$

$$I(Y_1, Y_2, \dots, Y_m) = D_{f_Y \| \tilde{f}_Y} = \int_{-\infty}^{\infty} f_Y(y) \log \left(\frac{f_Y(y)}{\prod_{i=1}^m \tilde{f}_{Y_i}(y_i)} \right) dY$$

$\tilde{f}_Y(y) = \prod_{i=1}^m \tilde{f}_{Y_i}(y_i), \tilde{f}_{Y_i}(y_i)$: Marginal p.d.f

- Learning Rule for ICA

$$\Delta w_{ik} = -\eta \frac{\partial}{\partial w_{ik}} D_{f \| \tilde{f}}$$

ICA(Independent Component Analysis)

- Kullback-Leibler Divergence

$$D_{f_Y \parallel \tilde{f}_Y} = \int_{-\infty}^{\infty} f_Y(y) \log \left(\frac{f_Y(y)}{\prod_{i=1}^m \tilde{f}_{Y_i}(y_i)} \right) dy$$

$$D_{f_Y \parallel \tilde{f}_Y} = \int_{-\infty}^{\infty} f_Y(y) \log f_Y(y) dy - \sum_{i=1}^m \int_{-\infty}^{\infty} f_Y(y) \log \tilde{f}_{Y_i}(y_i) dy$$

- The second term is

$$\begin{aligned} \int_{-\infty}^{\infty} \log \tilde{f}_{Y_i}(y_i) \left[\int_{-\infty}^{\infty} f_Y(y) dy^{(i)} \right] dy_i &= \int_{-\infty}^{\infty} \tilde{f}_{Y_i}(y_i) \log \tilde{f}_{Y_i}(y_i) dy_i \\ &= -\tilde{h}(Y_i) : \text{marginal entropy} \end{aligned}$$

- Kullback-Leibler Divergence

$$D_{f_Y \parallel \tilde{f}_Y} = -h(Y) + \sum_{i=1}^m \tilde{h}(Y_i)$$

ICA(Independent Component Analysis)

- Entropy $h(Y)$

$$h(Y) = h(WX) = h(X) + \log |\det(W)|,$$

$$(f_Y(y) = |\det(W)|^{-1} f_X(x), \quad dy = |\det(W)| dx)$$

- Marginal entropy $h(Y_i)$

Pdf of Y_i is obtained using truncate of Gram-Charlier series

$$\tilde{f}_{Y_i}(y_i(W)) = \alpha(y_i)[1 + \sum_{k=3}^{\infty} c_{ik} H_k(y_i)]$$

where

$$\alpha(y_i) = 1/\sqrt{2\pi} \exp(-y_i^2)$$

$H_k(y_i)$: Hermite polynomials

Cumulants $\{c_{ik} : k = 3, 4, \dots\}$ is obtained from k -th order moment of Y_i

Hermite polynomials: $H_3(y) = y^3 - 3y, H_4(y) = y^4 - 6y^2 + 3, \dots$

ICA(Independent Component Analysis)

- $\tilde{f}_{Y_i}(y_i(W)) = \alpha(y_i)[1 + \sum_{k=3}^{\infty} c_{ik} H_k(y_i)]$
- The index grouping is done as $k = (0), (3), (4,6), (5,7,9), \dots$
- By choosing by $k = (4,6)$

$$\tilde{f}_{Y_i}(y_i) = \alpha(y_i) \left(1 + \frac{k_{i,3}}{3!} H_3(y_i) + \frac{k_{i,4}^2}{4!} H_4(y_i) + \frac{(k_{i,6} + 10k_{i,3}^2)}{6!} H_6(y_i) \right)$$

- c_{ik} and k -th order moment of Y_i

$$k_{i,3} = m_{i,3}, \quad k_{i,4} = m_{i,4} - 3m_{i,2}^2$$

$$k_{i,6} = m_{i,6} - 10m_{i,3}^2 - 15m_{i,2}m_{i,4} + 30m_{i,2}^3$$

$$m_{i,k} = E[Y_i^k] = E \left[\left(\sum_{j=1}^m w_{ij} X_j \right)^k \right]$$

ICA(Independent Component Analysis)

- The cumulants are functions of W .
- Gradient of K-L divergence

$$\begin{aligned} 1) \frac{\partial}{\partial w_{ij}} \log(\det(W)) &= \frac{1}{\det(W)} \frac{\partial}{\partial w_{ij}} \det(W) \\ &= \frac{A_{ij}}{\det(W)} = (W^{-T})_{ij} \quad (\text{Adjugate matrix}) \end{aligned}$$

$$2) \frac{\partial \kappa_{i,3}}{\partial w_{ij}} \approx 3y_i^2 x_j, \quad \frac{\partial \kappa_{i,4}}{\partial w_{ij}} \approx -8y_i^3 x_j \dots\dots$$

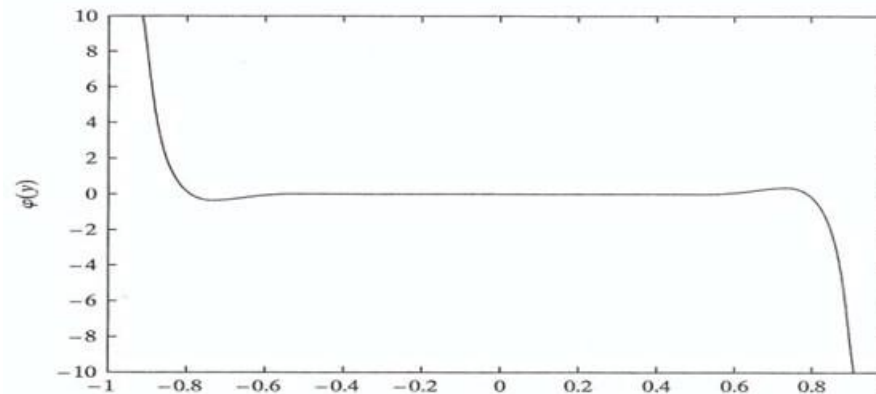
ICA(Independent Component Analysis)

- Minimization of Kullback-Leibler Divergence

$$D_{f_Y \| \tilde{f}_Y} = -h(Y) + \sum_{i=1}^m \tilde{h}(Y_i)$$

$$\frac{\partial}{\partial w_{ij}} D_{f \| \tilde{f}}(W) \approx -(W^{-T})_{ij} + \varphi(y_i)x_j$$

$$\varphi(y_i) = \frac{1}{2}y_i^5 + \frac{2}{3}y_i^7 + \frac{15}{2}y_i^9 + \frac{2}{15}y_i^{11} - \frac{112}{3}y_i^{13} + 128y_i^{15} - \frac{512}{3}y_i^{17}$$



ICA(Independent Component Analysis)

- Learning algorithm for ICA

$$\begin{aligned}\Delta w_{ij} &= -\eta \frac{\partial}{\partial w_{ij}} D_{f||f} \\ &= \eta \left((W^{-T})_{ij} - \phi(y_i) x_j \right)\end{aligned}$$

$$\Delta W = \eta (W^{-T} - \phi(y) \mathbf{x}^T)$$

$$\begin{aligned}\Delta W &= \eta [\mathbf{I} - \phi(y) \mathbf{x}^T W^T] W^{-T} \\ &= \eta [\mathbf{I} - \phi(y) \mathbf{y}^T] W^{-T}\end{aligned}$$

$$W(n+1) = W(n) + \eta(n) [\mathbf{I} - \phi(y(n)) \mathbf{y}^T(n)] W^{-T}(n)$$

ICA(Independent Component Analysis)

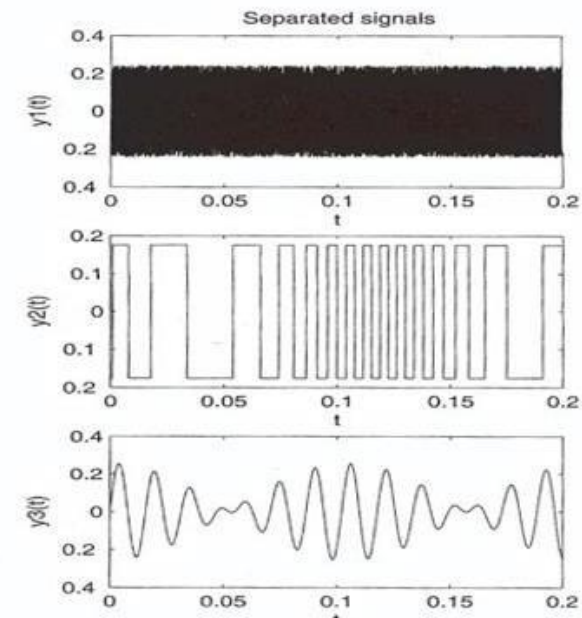
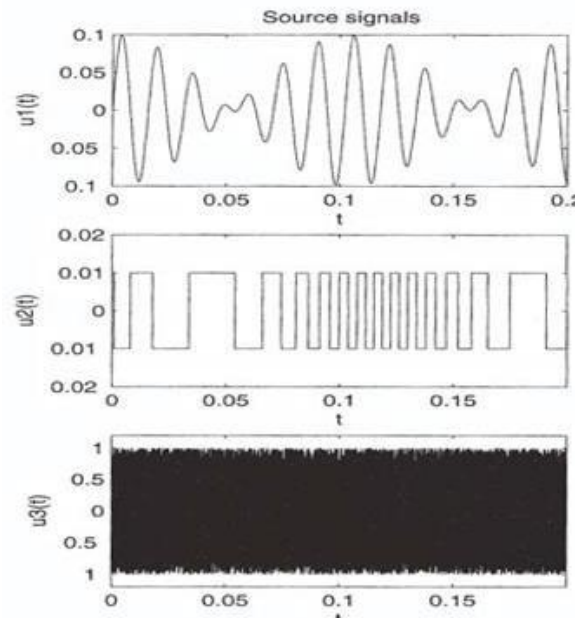
- Experiments

$$u_1(n) = 0.1\sin(400n)\cos(30n)$$

$$u_2(n) = 0.01 \operatorname{sgn}(\sin(500n + 9 \cos(40n)))$$

$$u_3(n) = \text{noise uniformly distributed in } [-1, 1]$$

$$A = \begin{bmatrix} 0.56 & 0.79 & -0.37 \\ -0.75 & 0.65 & 0.86 \\ 0.17 & 0.32 & -0.48 \end{bmatrix}$$



Exercise

- In computer science(CS) department, the probability of dropping the machine learning(ML) course in March is $1/6$, that in April is $1/3$, and the probability of taking ML course to the end without dropping is $1/2$, whereas those in Electrical engineering(EE) department are $1/8$, $1/8$, and $3/4$, respectively. Meanwhile, the portions of CS & EE students in ML course are $1/5$ & $4/5$, respectively. Letting X be the random variable on dropping or not of a student, and Y be the random variable on the department of a student, find the followings.
 1. Conditional entropy $H(X|Y)$.
 2. Mutual information $I(X; Y)$.

Exercise

- In computer science(CS) department, the probability of dropping the machine learning(ML) course in March is $1/6$, that in April is $1/3$, and the probability of taking ML course to the end without dropping is $1/2$, whereas those in Electrical engineering(EE) department are $1/8$, $1/8$, and $3/4$, respectively. Meanwhile, the portions of CS & EE students in ML course are $1/5$ & $4/5$, respectively. Letting X be the random variable on dropping or not of a student, and Y be the random variable on the department of a student, find $H(X|Y)$, $I(X; Y)$.
- 서술식을 수식으로 변경:

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$$H(X|Y) = H(X, Y) - H(Y).$$

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$$H(Y) = -\sum_{y \in Y} p(y) \log p(y) = -\frac{1}{5} \log \frac{1}{5} - \frac{4}{5} \log \frac{4}{5} = 0.7219$$

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$$H(X, Y) = -\sum_{x \in X} \sum_{y \in Y} p(x|y) \times p(y) \log p(x|y) \times p(y)$$

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$$= -\frac{1}{6} * \frac{1}{5} \log \left(\frac{1}{6} * \frac{1}{5} \right) - \frac{1}{3} * \frac{1}{5} \log \left(\frac{1}{3} * \frac{1}{5} \right) - \frac{1}{2} * \frac{1}{5} \log \left(\frac{1}{2} * \frac{1}{5} \right)$$

$$- \frac{1}{8} * \frac{4}{5} \log \left(\frac{1}{8} * \frac{4}{5} \right) - \frac{1}{8} * \frac{4}{5} \log \left(\frac{1}{8} * \frac{4}{5} \right) - \frac{3}{4} * \frac{4}{5} \log \left(\frac{3}{4} * \frac{4}{5} \right) = 1.8628$$

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$$- \frac{1}{8} * \frac{4}{5} \log \left(\frac{1}{8} * \frac{4}{5} \right) - \frac{1}{8} * \frac{4}{5} \log \left(\frac{1}{8} * \frac{4}{5} \right) - \frac{3}{4} * \frac{4}{5} \log \left(\frac{3}{4} * \frac{4}{5} \right) = 1.8628$$

$$\begin{aligned} H(X|Y) &= 1.8628 - 0.7219 \\ &= 1.1409 \end{aligned}$$

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 $I(X; Y) = H(X) + H(Y) - H(X, Y) = ?$

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By total probability,

$$P(X = x) = \sum_{y \in Y} P(X = x|Y = y)P(Y = y)$$

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$$H(X, Y) = 1.8628, H(Y) = 0.7219$$

$$H(X) = -\sum_{x \in X} p(x) \log p(x)$$

By total probability,

$$P(X = x) = \sum_{y \in Y} P(X = x|Y = y)P(Y = y)$$

$$P(X = 0) = \frac{1}{6} * \frac{1}{5} + \frac{1}{8} * \frac{4}{5} = \frac{2}{15}, P(X = 1) = \frac{1}{3} * \frac{1}{5} + \frac{1}{8} * \frac{4}{5} = \frac{1}{6}, P(X = 2) = \frac{1}{2} * \frac{1}{5} + \frac{3}{4} * \frac{4}{5} = \frac{7}{10}$$

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$$I(X; Y) = H(X) + H(Y) - H(X, Y) = ?$$

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$$H(X) = -\left[\frac{2}{15} \log\left(\frac{2}{15}\right) + \frac{1}{6} \log\left(\frac{1}{6}\right) + \frac{7}{10} \log\left(\frac{7}{10}\right) \right] = 1.1786$$

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$$I(X; Y) = H(X) + H(Y) - H(X, Y) = ?$$

$$H(X, Y) = 1.8628, H(Y) = 0.7219$$

$$H(X) = -\sum_{x \in X} p(x) \log p(x)$$

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$$H(X) = -\left[\frac{2}{15} \log\left(\frac{2}{15}\right) + \frac{1}{6} \log\left(\frac{1}{6}\right) + \frac{7}{10} \log\left(\frac{7}{10}\right) \right] = 1.1786$$

$$I(X, Y) = 1.1786 + 0.7219 - 1.8628 = 0.038$$

Summary

- Information
- Entropy
- Cross Entropy
- Error Backpropagation Learning
- Mutual Information
- Kullback Leibler Divergence
- Independent Component Analysis (ICA)
- Learning for ICA
- Blind Source Separation

Reference: Simon Haykin, *Neural Networks: A Comprehensive Foundation*, Prentice Hall